KUTAISI INTERNATIONAL UNIVERSITY

Scientific Computing

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1.5: Power series and Laplace transform

Outline

Lecture 1.5: Power series method and Laplace transform

- Power series method: recurrence for the coefficients; $y'' + y = 0$ and the Airy equation
- Laplace transform: definition, derivative rule, transform pairs
- Inverse transform by partial fractions
- Solving initial value problems with the Laplace transform
- Q & A

Series: 1.1 classification. 1.2 models. 1.3 existence and uniqueness. 1.4 scalar closed-form solutions. 1.5 power series and Laplace. 1.6 linear systems.

Power series method

Algorithm 1.1 (Power series method for a linear ODE with variable coefficients, centred at x_0)

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Remark 1.2 (The power series method covers many equations with variable coefficients, but the series may converge only in a limited interval)

The next example has an entire right-hand side but a series with radius of convergence 1.

Power series: the equation $y'' + y = 0$

Example 1.3 (Power series for $y'' + y = 0$ at $x_0 = 0$: $a_{n+2} = -\frac{a_n}{(n+2)(n+1)}$ gives $y = a_0 \cos x + a_1 \sin x$)

With $y = \sum a_n x^n$: $y'' = \sum_{n \geq 0} (n+2)(n+1) a_{n+2} x^n$. The equation becomes

$$\sum_{n=0}^{\infty} \left[(n+2)(n+1) a_{n+2} + a_n \right] x^n = 0 \implies a_{n+2} = -\frac{a_n}{(n+2)(n+1)}.$$

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Even index: $a_2 = -\frac{a_0}{2!}$, $a_4 = \frac{a_0}{4!}$, $a_6 = -\frac{a_0}{6!}, \ldots$ Odd index: $a_3 = -\frac{a_1}{3!}$, $a_5 = \frac{a_1}{5!}, \ldots$

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$$y = a_0\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\right) + a_1\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right) = a_0 \cos x + a_1 \sin x.$$

Power series: the Airy equation $y'' - xy = 0$

Example 1.4 (Power series for the Airy equation $y'' - xy = 0$ at $x_0 = 0$: $a_2 = 0$ and $a_{n+2} = \frac{a_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$)

$y'' = \sum_{n\geq 0}(n+2)(n+1)a_{n+2}x^n$ and $xy = \sum_{n\geq 1} a_{n-1}x^n$. Comparing coefficients of x^n:

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$a_3 = \frac{a_0}{3\cdot 2}$, $a_4 = \frac{a_1}{4\cdot 3}$, $a_5 = 0$, $a_6 = \frac{a_3}{6\cdot 5} = \frac{a_0}{180}$, $a_7 = \frac{a_4}{7\cdot 6} = \frac{a_1}{504}$.

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$$a_3 = \frac{a_0}{3\cdot 2}, \ a_4 = \frac{a_1}{4\cdot 3}, \ a_5 = 0, \ a_6 = \frac{a_3}{6\cdot 5} = \frac{a_0}{180}, \ a_7 = \frac{a_4}{7\cdot 6} = \frac{a_1}{504}.$$

$$y = a_0\left(1 + \frac{x^3}{6} + \frac{x^6}{180} + \cdots\right) + a_1\left(x + \frac{x^4}{12} + \frac{x^7}{504} + \cdots\right).$$

Power series can have a finite radius of convergence

Example 1.5 (The IVP $y' = y^2$, $y(0) = 1$ has the power series solution $\sum_{n\geq 0} x^n = \frac{1}{1-x}$ with radius of convergence 1)

$\sum a_n x^n$ with $a_0 = 1$: $\sum (n+1)a_{n+1}x^n = \left(\sum a_n x^n\right)^2$ gives $(n+1)a_{n+1} = \sum_{k=0}^{n} a_k a_{n-k}$, so $a_n = 1$ for all n by induction.

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Laplace transform: definition and basic rules

Definition 1.6 (Laplace transform $F(s) = \mathcal{L}\{f\}(s) = \int_0^\infty e^{-st} f(t)\,dt$ of a function f on $t \geq 0$)

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Proposition 1.7 (Rules: linearity, $\mathcal{L}\{f'\} = sF - f(0)$, $\mathcal{L}\{f''\} = s^2F - sf(0) - f'(0)$, and $\mathcal{L}\{e^{at}f\} = F(s-a)$)

Derivative rule: integrating by parts, $\int_0^\infty e^{-st} f'\,dt = \left[e^{-st} f\right]_0^\infty + s\int_0^\infty e^{-st} f\,dt = -f(0) + sF(s)$. *Applying it twice gives the rule for* f''. *The shift rule follows from* $\int_0^\infty e^{-st} e^{at} f\,dt = F(s-a)$.

Laplace transforms of the elementary functions

Proposition 1.8 (Transform pairs: $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$, $\mathcal{L}\{\sin \omega t\} = \frac{\omega}{s^2+\omega^2}$, $\mathcal{L}\{\cos \omega t\} = \frac{s}{s^2+\omega^2}$)

For $s > a$: $\int_0^\infty e^{-(s-a)t} dt = \dfrac{1}{s-a}$. Taking $a = \pm i\omega$ and using $e^{i\omega t} = \cos \omega t + i \sin \omega t$ gives the other two.

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Definition 1.9 (Inverse Laplace transform $\mathcal{L}^{-1}\{F\}$: the function f whose transform is F)

Methods: table lookup; partial fractions; the convolution theorem $\mathcal{L}^{-1}\{GH\} = g * h = \int_0^t g(\tau) h(t-\tau)\, d\tau$; the Bromwich integral $f(t) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} e^{st} F(s)\, ds$ with c to the right of all singularities of F (advanced).

Inverse Laplace transform by partial fractions

Example 1.10 ($\mathcal{L}^{-1}\left\{\frac{3s+1}{s^2+s-6}\right\} = \frac{8}{5}e^{-3t} + \frac{7}{5}e^{2t}$ by partial fractions)

$s^2 + s - 6 = (s+3)(s-2)$ and $\dfrac{3s+1}{(s+3)(s-2)} = \dfrac{A}{s+3} + \dfrac{B}{s-2}$, i.e.

$3s + 1 = A(s-2) + B(s+3)$.

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$$\mathcal{L}^{-1}\left\{\frac{8/5}{s+3} + \frac{7/5}{s-2}\right\} = \frac{8}{5}e^{-3t} + \frac{7}{5}e^{2t}.$$

Solving initial value problems with the Laplace transform

Algorithm 1.11 (Solving a linear IVP with constant coefficients by the Laplace transform)

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3. Solve the **algebraic** equation for $Y(s)$.
4. Invert: $y(t) = \mathcal{L}^{-1}\{Y(s)\}$.

Initial conditions are built in, and forcing terms such as sines, exponentials or piecewise functions are handled in the same way.

Laplace transform: worked example

Example 1.12 ($y'' + 4y = \sin 3t$, $y(0) = 1$, $y'(0) = 0$ has the solution $y = \cos 2t + \frac{3}{10}\sin 2t - \frac{1}{5}\sin 3t$)

Transform: $s^2 Y - s \cdot 1 - 0 + 4Y = \dfrac{3}{s^2+9}$, so $Y = \dfrac{s}{s^2+4} + \dfrac{3}{(s^2+4)(s^2+9)}$.

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Partial fractions: $\dfrac{3}{(s^2+4)(s^2+9)} = \dfrac{3}{5}\left(\dfrac{1}{s^2+4} - \dfrac{1}{s^2+9}\right)$, hence

$$Y = \frac{s}{s^2+4} + \frac{3}{10}\cdot\frac{2}{s^2+4} - \frac{1}{5}\cdot\frac{3}{s^2+9}.$$

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Invert with the table: $y(t) = \cos 2t + \dfrac{3}{10}\sin 2t - \dfrac{1}{5}\sin 3t$. Check: $y(0) = 1$, $y'(0) = \frac{3}{5} - \frac{3}{5} = 0$.

References and further reading

- Power-series methods: J. Cano, S. Falkensteiner, J. R. Sendra, *Existence and convergence of Puiseux series solutions for autonomous first-order differential equations*, `arXiv:1908.09196` (2019).
- Laplace transform for IVPs: MIT OpenCourseWare 18.03 *Differential Equations*, Laplace transform chapter (`ocw.mit.edu`); Boyce, DiPrima & Wilson, *Elementary Differential Equations and Boundary Value Problems*.
- All worked examples are verified symbolically by `closed_form_check.py`.

Q & A